

HTTP Web Server in Java - Project Report

1. Project Overview

This project explains the development of a simple HTTP Web Server using Java Socket Programming. The server listens for client requests through a browser and sends HTTP responses containing web pages.

2. Objective

The main objective of this project is to understand the working of HTTP protocol, client-server communication, and web server implementation using Java.

3. Technologies Used

Programming Language: Java

Concepts Used: Socket Programming, HTTP Protocol

Development Tool: IntelliJ IDEA / Eclipse

Browser: Google Chrome / Firefox

4. System Architecture

The browser sends an HTTP request to the Java server. The server processes the request and sends an HTTP response containing HTML content back to the client browser.

5. Features of HTTP Web Server

- Handles HTTP Requests and Responses
- Uses Socket Programming
- Displays Web Pages in Browser
- Lightweight and Simple Design
- Supports Client-Server Communication

6. Working Procedure

1. Create a ServerSocket on a specific port.
2. Wait for client connection requests.
3. Read HTTP request from browser.
4. Generate HTTP response with HTML content.
5. Send response to client browser.
6. Close socket connection.

7. Sample Java Code

```
import java.io.*;
import java.net.*;
```

```

public class HttpWebServer {
    public static void main(String[] args) throws Exception {
        ServerSocket serverSocket = new ServerSocket(8080);
        while(true) {
            Socket socket = serverSocket.accept();
            BufferedReader in = new BufferedReader(
                new InputStreamReader(socket.getInputStream()));
            OutputStream out = socket.getOutputStream();
            String html = "<h1>HTTP Web Server in Java</h1>";
            String response =
                "HTTP/1.1 200 OK\r\n" +
                "Content-Type: text/html\r\n\r\n" +
                html;
            out.write(response.getBytes());
            socket.close();
        }
    }
}

```

8. Advantages

- Easy to understand HTTP communication
- Useful for networking projects
- Helps learn web server architecture
- Lightweight implementation

9. Applications

- Educational Projects
- Networking Demonstrations
- Web Server Fundamentals
- Java Socket Programming Practice

10. Future Enhancements

- Add multi-threading support
- Support multiple web pages
- Implement file handling
- Add database connectivity
- Improve security features

11. Conclusion

The HTTP Web Server project demonstrates the practical implementation of Java networking concepts and HTTP communication. It helps understand how web servers process client requests and deliver responses.

12. Technology Summary Table

Module	Technology
Programming Language	Java
Networking	Socket Programming
Protocol	HTTP
IDE	Eclipse / IntelliJ IDEA